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Questions

Q1.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The pancreas is an organ with many functions.

A student observed some cells of the pancreas using an electron microscope.

The student described one organelle that was seen as

"several curved membrane-bound sacs of decreasing size".

(a) (i) Which organelle was seen?

(1)

☐ **A** centriole

☐ **B** Golgi apparatus

☐ **C** rough endoplasmic reticulum

☐ **D** smooth endoplasmic reticulum

(ii) Explain why this organelle cannot be seen using a light microscope.

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(iii) Give **two** differences between an organ and a tissue.

(2)

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(b) Plant cells can also be studied using an electron microscope.

Name **two** structures present in animal cells that are not present in a plant cell.

(2)

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(c) Cells in the pancreas use amino acids to synthesise polypeptides.

These cells also secrete enzymes and glycoproteins into the small intestine.

Cells from the pancreas, containing amino acids labelled with a fluorescent marker, were used in an investigation. The fluorescent marker looked green when seen with a special microscope.

The percentage of green fluorescence inside and outside the cells was measured at the start and after 60 minutes.

The results of this investigation are shown in the table.

Time / min	Percentage of green fluorescence (%)	
	Inside the cells	Outside the cells
0	100	0
60	38	62

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Explain the results of this investigation.

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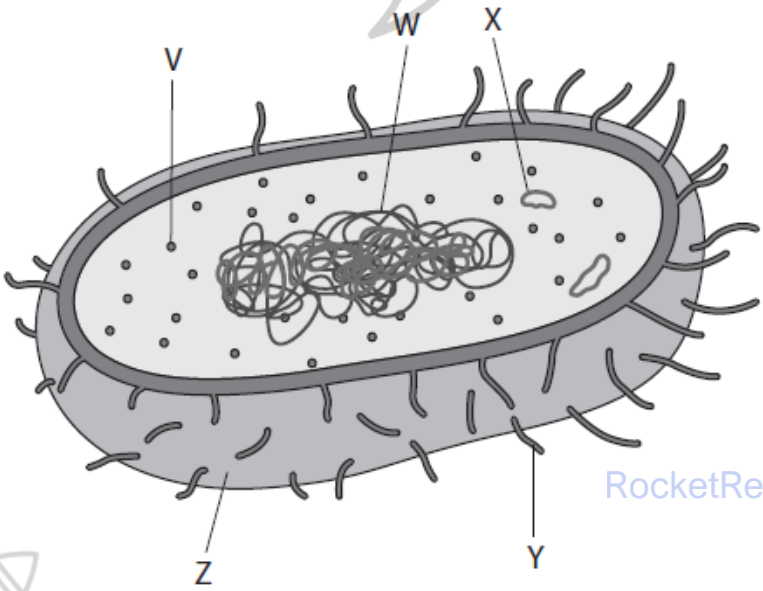
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(Total for question = 11 marks)

Q2.

The diagram shows a prokaryotic cell, drawn by a student.



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(a) (i) Which labelled structure is a plasmid?

(1)

☒ **A** W

☒ **B** X

☒ **C** Y

☒ **D** Z

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(ii) Which labelled structure enables this cell to adhere to other bacterial cells in conjugation?

(1)

☒ **A** V

☒ **B** X

☒ **C** Y

☒ **D** Z

(iii) State how ribosomes in prokaryotic cells are different from those in eukaryotic cells.

(1)

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(b) Prokaryotic cells may also have a capsule and a flagellum.

State **one** function of each of these structures.

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Capsule

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Flagellum

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Q3.

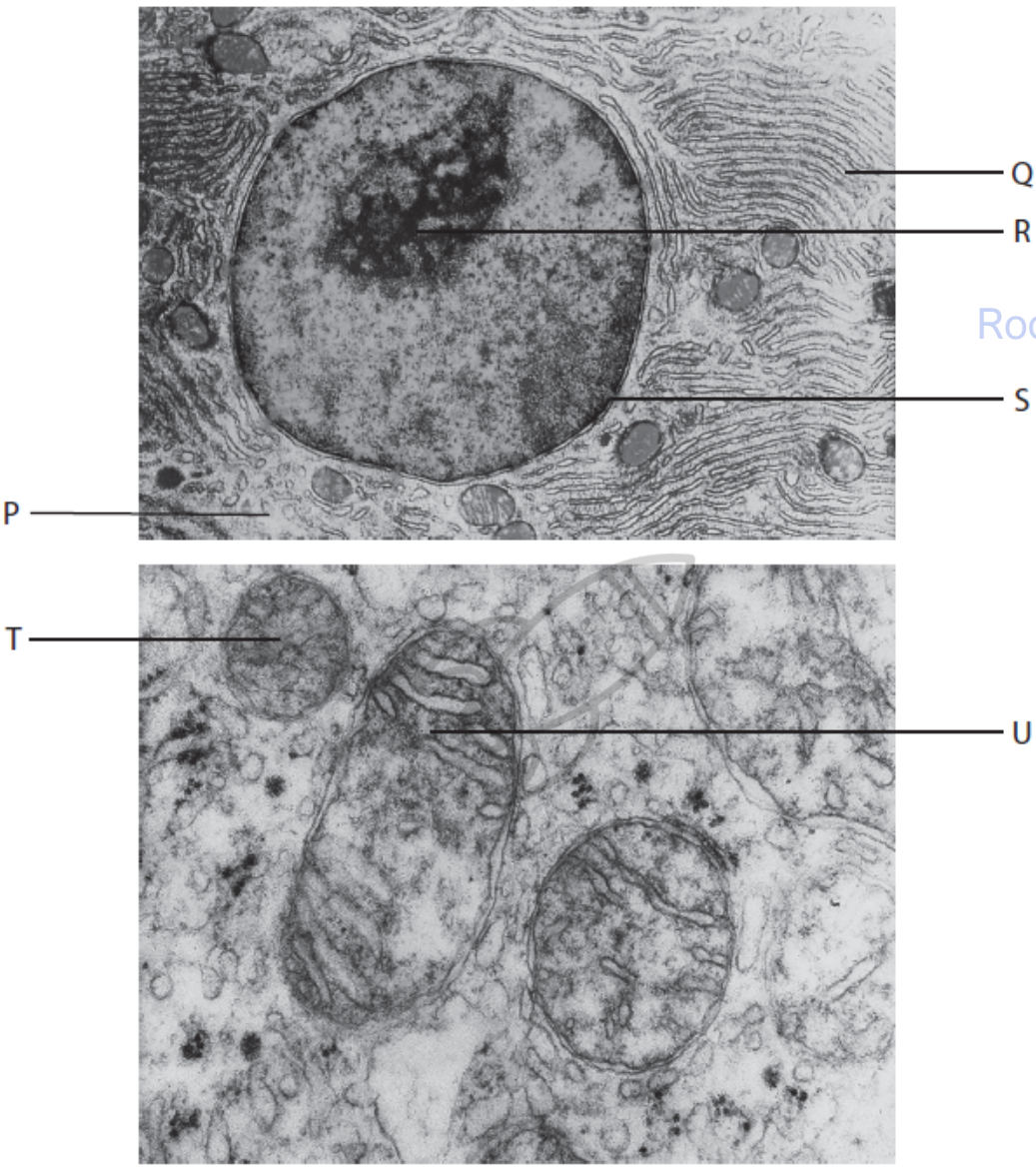
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Insulin is a protein produced in beta cells in the pancreas.

Ribosomes are involved in the production of insulin.

(a) The photographs show part of a beta cell from the pancreas, as seen using an electron microscope.



(Source: © Don Fawcett/Science Photo Library
© Martin M. Rotker/Science Photo Library)

(i) Which labelled structure produces ribosomes?

(1)

☒ **A** P

☒ **B** Q

☒ **C** R

☒ **D** S

(ii) The structures labelled T and U are the same type of organelle.

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Suggest why these structures look different in this photograph.

(2)

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(iii) A student measured the length of the structure labelled U as 52 mm.

The actual length of this structure is 3.0 µm.

Calculate the magnification of this electron microscope image.

(2)

Answer .....

(b) Describe how the protein insulin is transported from the ribosomes and then secreted by a beta cell.

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(c) Explain why beta cells can produce insulin but other cells in the pancreas do not.

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(Total for question = 15 marks)

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Q4.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

The number and size of Golgi apparatus vary depending on the type of cell.

(a) (i) Draw a labelled diagram of the Golgi apparatus.

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(4)

(ii) How many of the following are functions of the Golgi apparatus?

- formation of extracellular enzymes
- modification of proteins
- formation of peptide bonds through condensation reactions

(1)

☒ **A** none

☒ **B** one

☒ **C** two

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☒ **D** three

(iii) A cell was supplied with radioactive amino acids. The cell took in these amino acids and used them in protein synthesis.

Which structure in the cell would become radioactive first?

(1)

☒ **A** centrioles

☒ **B** Golgi apparatus

☒ **C** lysosomes

☒ **D** ribosomes

(b) In a mammalian cell, there was one Golgi apparatus with a diameter of 1.1 µm.

In a maize cell, there were 600 Golgi apparatus. The largest diameter of Golgi apparatus in the maize cell was 0.5 µm.

(i) Calculate the percentage difference in these diameters.

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(1)

Answer ..... %

(ii) The number and size of the Golgi apparatus in the maize cell increases during one stage of the cell cycle.

In which stage of the cell cycle would the number and size of the Golgi apparatus increase?

(1)

☒ **A** anaphase

☒ **B** interphase

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☒ **C** prophase

☒ **D** telophase

(iii) Suggest why the number and size of the Golgi apparatus change during the cell cycle.

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(Total for question = 12 marks)

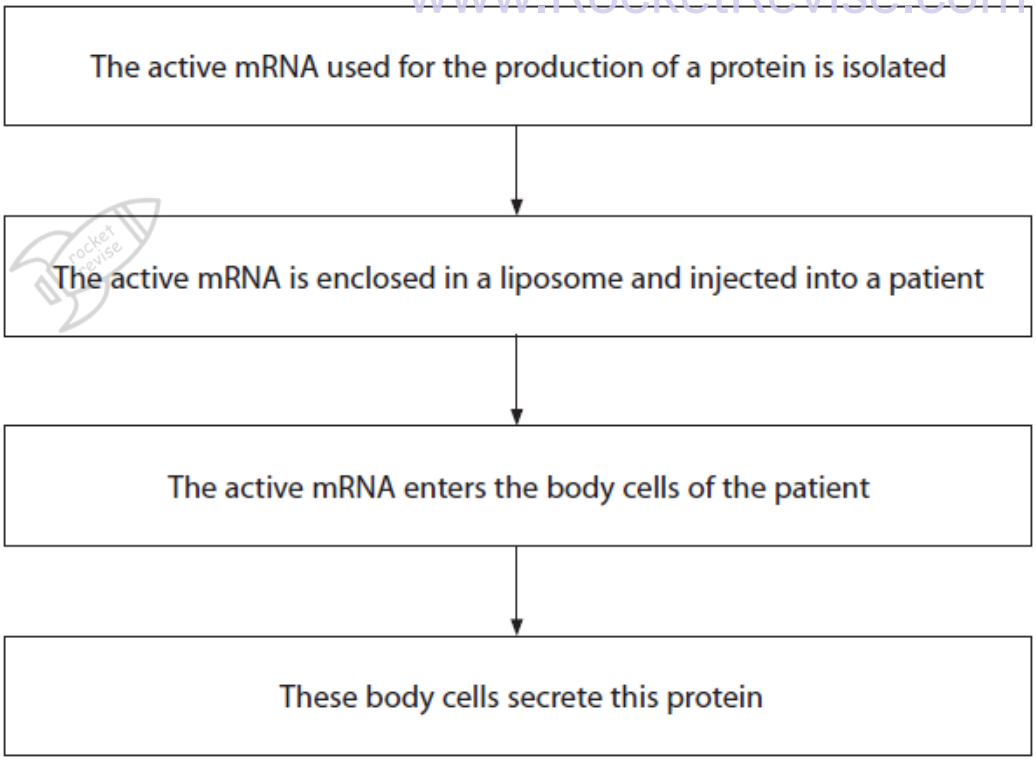
Q5.

A new vaccine has been developed that contains active mRNA.

The flow diagram shows how this vaccine causes body cells to produce a new protein.

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(a) The diameter of one spherical liposome was 75 nm.

Calculate the volume of this liposome using the formula:

$$V = \frac{4}{3}\pi r^3$$

Give your answer in standard form.

(2)

Answer ..... nm^3

(b) Active mRNA is formed after post-transcriptional modification of pre-mRNA.

The structure of pre-mRNA produced from transcription is different from the structure of active mRNA.

Compare and contrast the structures of pre-mRNA and active mRNA.

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(Total for question = 6 marks)

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Q6.

Red blood cells are produced from pluripotent stem cells found in bone marrow.

(a) Which statement about these stem cells is correct?

(1)

☒ **A** they can produce all types of cell

☒ **B** they can produce all types of cell except extraembryonic cells

☒ **C** they can produce some types of cell

☒ **D** they can produce red blood cells only

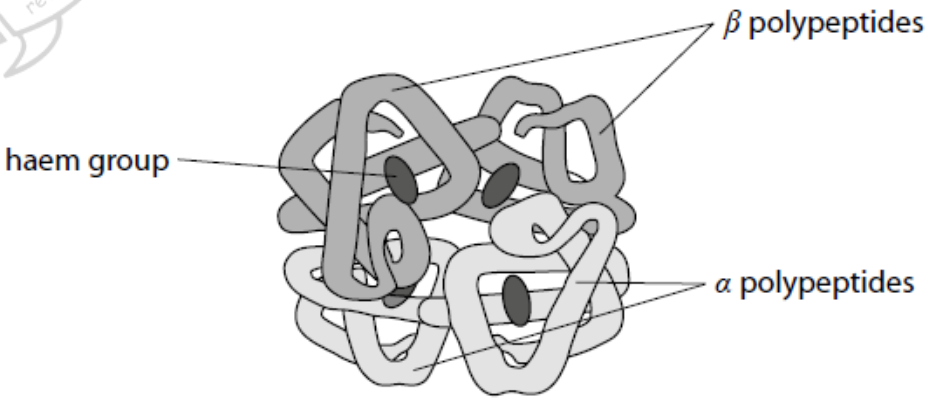
(b) Red blood cells contain haemoglobin.

A molecule of haemoglobin is made of four polypeptides. Each polypeptide has a haem group attached to it. The haem group is **not** made of amino acids.

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In most adult haemoglobin, there are two α polypeptides and two β polypeptides.

The diagram shows the structure of adult haemoglobin.



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Describe the role of the rough endoplasmic reticulum in the synthesis of haemoglobin.

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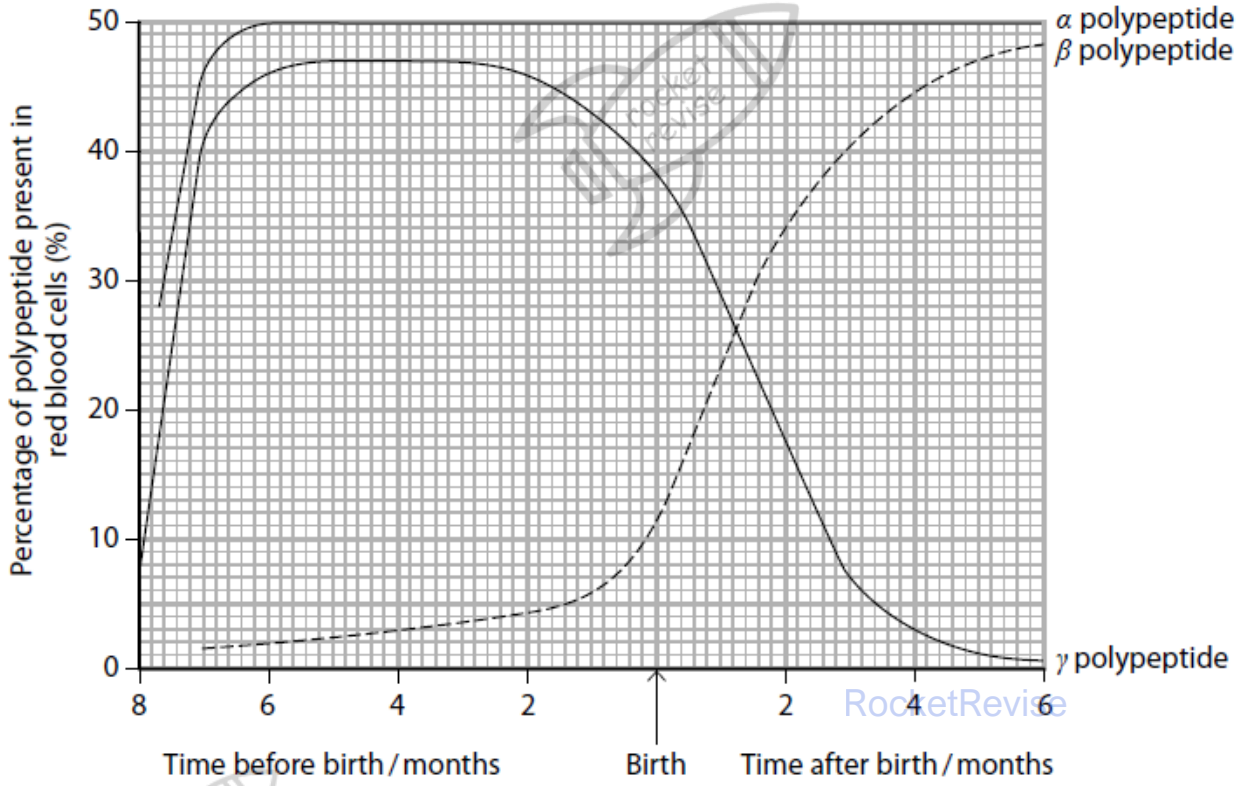
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(c) Fetal haemoglobin has a similar structure to adult haemoglobin. Fetal haemoglobin has two α polypeptides and two γ polypeptides.

The graph shows the percentage of each polypeptide present in red blood cells in an individual before and after birth.



(i) Describe the changes in the percentages of polypeptides present in red blood cells. Use the information in the graph to support your answer.

(3)

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(ii) Explain how epigenetic modification could result in these changes.

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(Total for question = 11 marks)

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Q7.

The photograph shows a grasshopper being caught in a web by a wasp spider.

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(Source: © Nicola Ferrari / Alamy Stock Photo)

The spider uses extracellular enzymes from its digestive system to digest parts of the grasshopper.

(a) The digestive system is an organ system.

State what is meant by the term **organ system.**

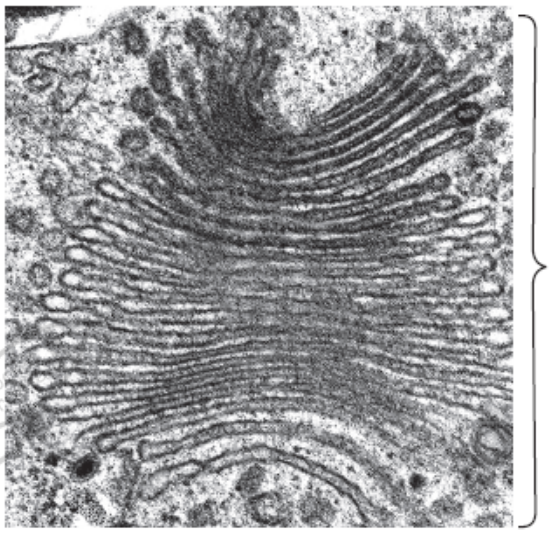
(1)

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(b) The photograph shows an organelle involved in the formation of extracellular enzymes, as seen using an electron microscope.



(Source: © Science Photo Library)

(i) Name organelle A, shown in the photograph.

(1)

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(ii) Describe the role of **this** organelle in the formation and transport of extracellular enzymes.

(2)

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(iii) A student measured the width of one part of the organelle in the photograph as 3.2 cm.

The actual width was 0.95 µm.

Calculate the magnification of the photograph.

Give your answer to **two** significant figures.

(2)

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(Total for question = 6 marks)

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